Fabrication Manual V2: The Grand Voyager (Hybrid Edition)

This revised manual updates the **Zenith Aero Bamboo Pedicab** build plan to match the **Long-Wheelbase (LWB)** aesthetic of the original artistic concept, while integrating **Carbon Fiber (CF) Reinforcement** for extreme structural performance.

1. Updated Chassis Proportions: "The Grand Voyager"

To replicate the "island" seat effect and spaciousness of the original concept , the frame has been elongated.

- **Total Vehicle Length**: 3050 mm (120")
- **Wheelbase (Axle-to-Axle)**: 1850 mm (72.8")
- **Main Chassis Rail (BB-01)**: 2700 mm (106.3")
- **Passenger Legroom**: 800 mm (31.5")

2. Hybrid Lamination & Carbon Fiber Schedule

The inclusion of Carbon Fiber allows for a thinner profile with significantly higher tensile strength and stiffness in the LWB frame.

Section	Bamboo Layers (5mm)	CF Reinforcement	Total Thickness
Main Chassis Rails	12 Layers	2 Layers (Internal Core)	65 mm
Main Joints	10 Layers	4 Layers (Sleeve Wrap)	60 mm
Wheel Arches	8 Layers	2 Layers (Outer Skin)	45 mm

Carbon Fiber Application:

- **Core Reinforcement**: Place Uni-Directional (UD) carbon fiber strips between the 6th and 7th layers of bamboo during the lamination process.
- **Joint Sleeving**: After the "overbuilt" bamboo tongue is ground to shape, wrap the joint in 2-4 layers of 3K Twill Carbon Fiber cloth before final epoxy coating.

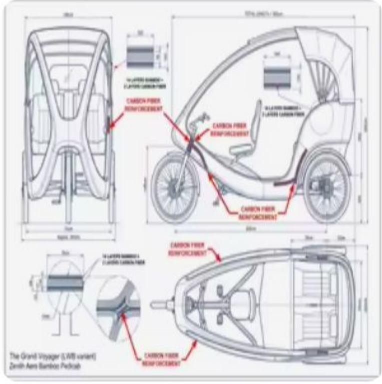
3. Engineering Detail: Hybrid Bridle Joint

The "Grand Voyager" uses a hybrid reinforcement strategy at the critical chassis-to-axle transition.

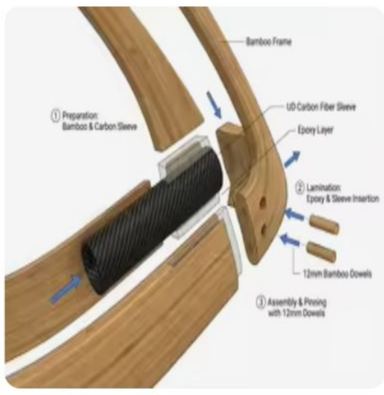
- **CF Core**: Acts as a "spine," preventing the 2700mm rail from flexing under load.
- **Dowel Locking**: 12mm bamboo dowels must pass through the CF layers. Use a carbide-tipped drill bit to prevent splintering the carbon fiber.
- **Epoxy Selection**: Use a high-modulus, slow-cure structural resin (e.g., Entropy Resins Super Sap) which is compatible with both cellulose (bamboo) and carbon fibers.

4. Technical Diagrams (V2)

Grand Voyager Proportions & LWB Views



Hybrid Bamboo-Carbon Joint Detail



5. Collaboration & Academic Alignment

The transition to a hybrid bamboo-carbon construction aligns with high-performance research often seen at institutions like **UBC**. This combination offers:

1. **Vibration Damping**: Bamboo absorbs high-frequency road buzz.
2. **Structural Rigidity**: Carbon fiber provides the low-flex platform required for a 3-meter chassis.
3. **Sustainability**: Minimizes carbon fiber usage to critical "stress-only" zones while utilizing bamboo for 95% of the volume.

Design Credit: This hybrid concept was co-developed by **Accio.ai** for the Zenith Aero series.